Metric3D v2

A Versatile Monocular Geometric Foundation Model for Zero-Shot Metric Depth and Surface Normal Estimation

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What Can You Measure from a Single Photo?

Goal: From one RGB image, recover per-pixel

- **Metric depth** — distance in meters from camera to each point
- **Surface normals** — 3D orientation of each surface patch

Demo output:
RGB | Depth | Normals
(replace with inference result)

Connection to course: The pinhole camera model (Ch. 2) relates 3D world coordinates to 2D image coordinates via focal length f:

$$u = f\,\frac{X}{Z}, \qquad v = f\,\frac{Y}{Z}$$

Without knowing f, depth is ambiguous.

Why Is This Hard? — Metric Ambiguity

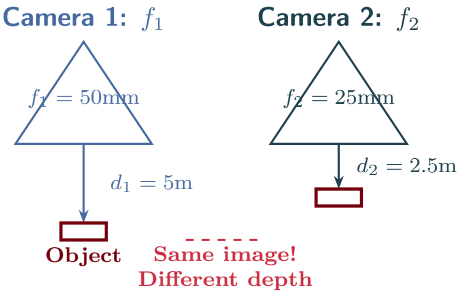
Key insight: Two cameras with different focal lengths can produce *identical* images of a scene at different depths.

The apparent depth scales with focal length:

$$\boxed{d_a = \hat{S} \cdot \frac{\hat{f}}{\hat{S}'}}$$

where $\hat{S}$ is real-world size, $\hat{f}$ is focal length, and $\hat{S}'$ is projected size.

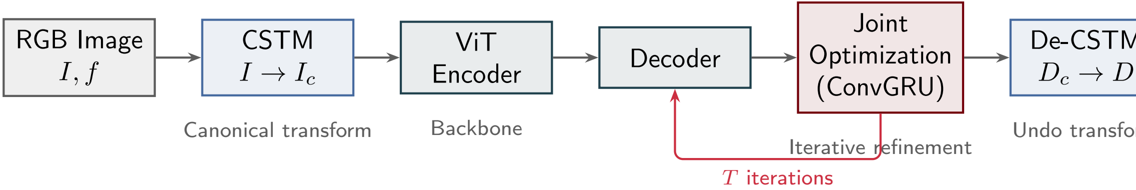
A network trained on mixed datasets with different cameras sees **conflicting depth labels** for visually identical patches.



Depth Estimation Taxonomy

Metric3D v2 achieves *zero-shot metric depth* by explicitly modeling camera intrinsics, unlike methods that learn only relative ordering.

Architecture Overview



- **Modular:** Any backbone (ViT-S, ViT-L, ViT-G) plugs in; CSTM is backbone-agnostic
- **Two key innovations:** Canonical Camera Space Transform (CSTM) and Joint Depth-Normal Optimization

Focal Length Is Everything

The paper establishes two critical observations:

O1: Sensor/pixel size does not matter.
Only the ratio of focal length to pixel size matters, and this is captured by the focal length in pixels f_{px}.

O2: Focal length is vital for metric depth.
Different focal lengths cause different depth scales. The network must compensate, or depth predictions collapse across diverse cameras.

Normals are metric-agnostic:
Surface normals depend only on local geometry, not on absolute depth scale. This

$$D = f \cdot \frac{B}{d}$$

Depth $\propto$ focal length

Needs camera info

$$\mathbf{n} = \frac{\partial \mathbf{P}}{\partial u} \times \frac{\partial \mathbf{P}}{\partial v}$$

Normals: local geometry only

Camera-agnostic

Canonical Camera Space Transform — Label Space

Idea: Map all training data to a "canonical" camera with fixed focal length $f^c = 1000\,\text{px}$.

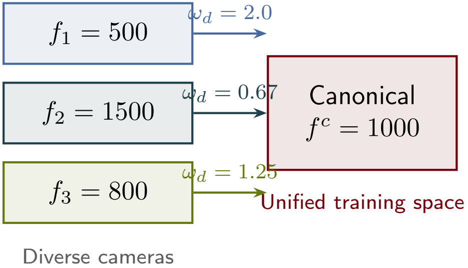
CSTM$_{\textbf{label}}$ transforms depth annotations:

$$\boxed{D_c^* = \omega_d \cdot D^*, \qquad \omega_d = \frac{f^c}{f}}$$

If $f < f^c$: depth is *scaled up* (wide-angle cameras see closer).
If $f > f^c$: depth is *scaled down*.

At inference, **De-CSTM** reverses:

$$D = \frac{D_c}{\omega_d} = D_c \cdot \frac{f}{f^c}$$



Canonical Camera Space Transform — Image Space

CSTM$_{\text{image}}$ resizes the input image instead of scaling labels:

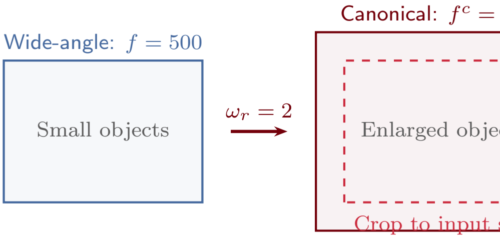
$$\boxed{I_c = \mathcal{T}(I,\ \omega_r), \qquad \omega_r = \frac{f^c}{f}}$$

Why resize? After resizing, objects in I_c appear at the angular size they would subtend through the canonical camera. The encoder sees a consistent "canonical viewpoint."

Both paths are equivalent:

- CSTM$_{\text{label}}$: scale depth labels, keep images
- CSTM$_{\text{image}}$: resize images, keep depth labels

In practice, the paper uses **CSTM$_{\text{l}}$**



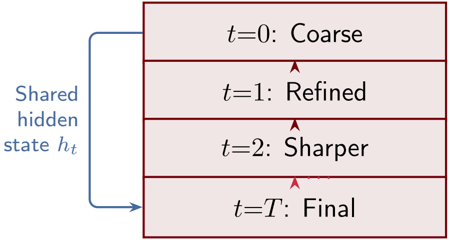
Joint Depth-Normal Optimization

Novel contribution: First method to jointly optimize metric depth and surface normals through an iterative process.

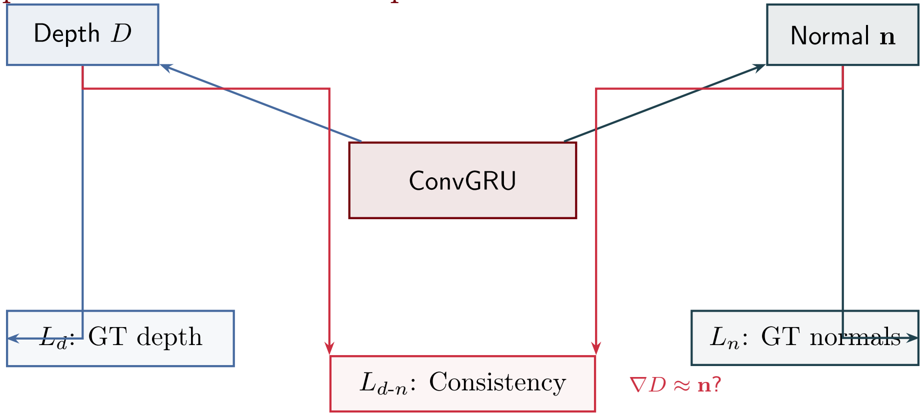
ConvGRU iterative refinement:

1. Decoder produces initial coarse estimates D_0, $\mathbf{n}_0$
2. At each iteration t:
   - Compute update Δ_t from hidden state
   - $D_{t+1} = D_t + \Delta_{D,t}$
   - $\mathbf{n}_{t+1} = \mathbf{n}_t + \Delta_{\mathbf{n},t}$
3. After T steps: final refined output

Why jointly? Depth and normals are geometrically coupled. Sharing a hidden state lets each task inform the other.



Joint Optimization — Three Supervision Sources



Converts depth gradients to pseudo-normals; penalizes disagreement

Key insight: Only ∼20K outdoor images have GT normals. The consistency loss $L_{d\text{-}n}$ lets the model learn normals from depth-only datasets by enforcing geometric agreement.

Random Proposal Normalization Loss (RPNL)

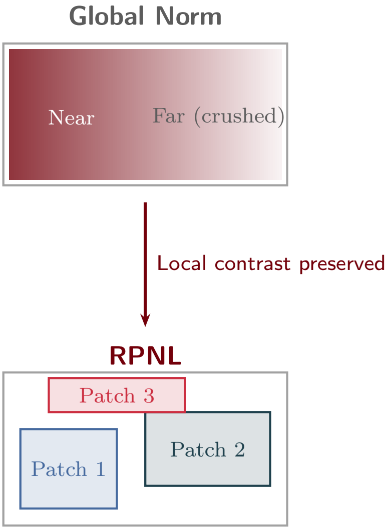
Problem: Standard depth losses normalize globally. Close objects dominate; distant detail is lost.

Analogy from course: This is the same problem *adaptive histogram equalization* solves for image contrast. Global normalization washes out local structure.

RPNL solution:

1. Randomly crop K patches from the depth map
2. Normalize depth within each patch independently
3. Compute loss per patch, then average

This forces the network to preserve *local*



Loss Function Summary

Total loss:

$$L = \underbrace{0.5\,L_d}_{\text{depth}} + \underbrace{1.0\,L_n}_{\text{normals}} + \underbrace{0.01\,L_{d\text{-}n}}_{\text{consistency}}$$

$$L_d = L_{\text{PWN}} + L_{\text{VNL}} + L_{\text{silog}} + L_{\text{RPNL}}$$

Pair-wise normal | Virtual normal | Scale-invariant log | Random proposal

$$L_n = 1 - \langle\, \hat{\mathbf{n}},\ \mathbf{n}^* \rangle \qquad \text{(cosine similarity)}$$

$L_{d\text{-}n}$: penalizes disagreement between ∇D (pseudo-normals from depth) and predicted $\mathbf{n}$

Note: $L_{d\text{-}n}$ weight is small (0.01) because pseudo-normals from predicted depth

Training at Scale

16M
images

16
datasets

10K+
camera models

48 $\times$ A100
GPUs

Indoor: NYUv2, ScanNet, Taskonomy, ...
Outdoor: KITTI, DDAD, Waymo, Lyft, ...
Mixed: DIML, 3D Ken Burns, BlendedMVS, ...

Critical data gap: Only $\sim$20K outdoor images have GT surface normals (mostly from ScanNet indoor). This motivates the depth-normal consistency loss $L_{d\text{-}n}$ as a way to learn normals from depth-only outdoor data.

Benchmark Results

Zero-shot metric depth (no fine-tuning on target dataset):

Dataset	**AbsRel** $\downarrow$	δ_1 $\uparrow$
NYUv2	**0.063**	**0.975**
KITTI	**0.043**	**0.989**
ScanNet	0.069	0.970
DIODE Out	0.308	0.832

Key: Zero-shot results that *beat* methods fine-tuned on the target dataset. CSTM enables generalization across camera types.

Demo output:

Our inference results on test images

RGB | Depth | Normals

(replace with inference result)

Downstream Applications

3D Reconstruction

Back-project depth to point cloud:

$$X = \frac{(u - u_0) \cdot D}{f}$$

$$Y = \frac{(v - v_0) \cdot D}{f}$$

$$Z = D$$

Table 6: competitive with MVS methods using a *single* image.

SLAM Improvement

Using Metric3D v2 as depth prior in visual SLAM:

- 10× reduction in drift (Table 7)
- Enables metric-scale monocular SLAM

Our Point Cloud Demo

Point cloud rendering

(replace with PLY render)

Strengths

1. **CSTM is elegant and modular.** A single scaling factor $\omega = f^c/f$ handles all camera diversity. No architecture changes needed; works with any backbone.
2. **Learning normals from depth-only data.** The consistency loss $L_{d\text{-}n}$ sidesteps the scarcity of outdoor normal annotations ($\sim$20K images). The network learns normals as a *byproduct* of depth prediction.
3. **Comprehensive evaluation.** Tested on 16+ benchmarks spanning indoor, outdoor, in-the-wild, and downstream tasks (3D reconstruction, SLAM, metrology).
4. **Immediately usable.** Available via PyTorch Hub with multiple model sizes (ViT-S, ViT-L, ViT-G). One line of code to run inference.

Limitations and Open Questions

1. **Requires camera intrinsics at inference.** Unlike UniDepth (CVPR 2024) and Depth Pro (Apple, 2024), which estimate intrinsics internally. Users must provide or estimate f.
2. **Single-frame only.** No temporal consistency for video. Sequential frames can produce flickering depth maps.
3. **Outdoor normal data scarcity persists.** The consistency loss helps but does not fully close the gap; outdoor normal quality lags indoor.
4. **Canonical focal length $f^c = 1000$ is ad hoc.** The paper provides no theoretical justification for this choice. Sensitivity analysis would strengthen the claim.
5. **Newer competitors are closing the gap.**
   - **MoGe** (CVPR 2025): jointly predicts 3D points + normals + FOV
   - **Depth Pro**: metric depth without intrinsics at 0.3s per image

Key Takeaways

1. CSTM — canonical camera transform

2. Joint optimization — depth + normals

3. RPNL — local depth normalization

Big idea: Transforming the *data space* (CSTM) is simpler and more effective than encoding camera parameters into the network architecture.

Geometric foundation models trained on massive diverse data can rival — and often beat — task-specific models fine-tuned on individual benchmarks.

Questions?